

Machine Learning and Regression Models for Stability Assessment of Cantilever Retaining Walls-Supplementary Material Manuscript

This file includes auxiliary information absent from the article titled "Machine Learning and Regression Models for Stability Assessment of Cantilever Retaining Walls," such as additional tables, figures, extensive formulas, and comprehensive result lists, which are essential for a complete understanding of the study, as well as for transparency and repeatability.

Three different datasets are analyzed to examine the impact of varying wall dimensions on failure scenarios. The first dataset, henceforth referred to as the *regular* set (E1), is taken from previous studies [1]. In this dataset, each design parameter has four levels, given by (X_1 : 0.25H, 0.50H, 0.75H, 1.0H), (X_2 : 0.15 X_1 , 0.30 X_1 , 0.45 X_1 , 0.60 X_1), (X_3 : 0.06H, 0.09H, 0.12H, 0.15H), (X_4 : 0, 0.01, 0.02, 0.03), and (ϕ : 20°, 27°, 34°, 41°). The combinations of these design parameters correspond to ($4^5 = 1024$) different CRW designs. The second dataset (henceforth, *random*, or E2), comprising 1,251 different CRW designs, is generated randomly to ensure that it differs from the 1,024 wall combinations in the regular dataset while remaining within the minimum and maximum limits of the design parameters. The third dataset (henceforth, *mixed*, or E3) is obtained by combining E1 and E2. For every member of these three datasets, the safety factors (F_s), (F_o), and (F_{ss}) are computed using the methods described earlier.

Table 1. Output statistics for the selected input datasets.

Dataset type	Safety Factor	#Wall Designs	Mean	Min	Max	STD	Med	CK	CS
Regular	F_s	1024	1.2916	0.1088	6.6784	1.1742	0.8529	3.1515	1.7397
Regular	F_o	1024	3.3735	0.2256	12.9331	2.9320	2.6081	0.6270	0.0917
Regular	F_{ss}	1024	1.5252	0.7000	2.9100	0.5219	1.4700	-0.6961	0.4199
Random	F_s	1251	1.2057	0.1698	4.9674	0.8748	0.8779	0.6659	1.1092
Random	F_o	1251	3.1559	0.3618	10.5725	2.2700	2.5486	0.2563	0.9878
Random	F_{ss}	1251	1.5089	0.7700	2.6700	0.4672	1.4900	-1.2964	0.2467

Dataset type	Safety Factor	#Wall Designs	Mean	Min	Max	STD	Med	CK	CS
Mixed	F_s	2275	1.2443	0.1088	6.6784	1.0214	0.8702	3.0841	1.6085
Mixed	F_o	2275	3.2538	0.2256	12.9331	2.5912	2.5718	0.8136	1.1083
Mixed	F_{ss}	2275	1.5162	0.7000	2.9100	0.4926	1.4700	-0.9257	0.3456

Min: minimum, Max: maximum, STD: standard deviation, CK: kurtosis, CS: skewness coefficients

1. Performance Evaluation Metrics of Machine Learning Models

The scalar model performance metrics used in the present study are listed in Table 2. They are the maximum error (MaxE), mean absolute error (MAE), mean absolute percentage error (MAPE), mean squared logarithmic error (MSLE), median absolute error (MedAE), Tweedie mean squared error (TMSE), Tweedie mean Poisson deviance (TMPE), Tweedie mean gamma deviance (TMGE), D^2 Tweedie score (D^2T), and D^2 pinball score (D^2P) [2, 3].

Table 2. Evaluation metrics for regression models.

Metric	Description	Metric	Description
MaxE	Maximum absolute error $\max_i y_{a(i)} - y_{p(i)} $	MPD	Mean Poisson Deviance (Tweedie with $p = 1$) $\frac{1}{n} \sum_{i=1}^n 2 \left(y_{a(i)} \ln \left(\frac{y_{a(i)}}{y_{p(i)}} \right) + y_{p(i)} - y_{a(i)} \right)$
MAE	Mean Absolute Error $\frac{1}{n} \sum_{i=1}^n y_{a(i)} - y_{p(i)} $	MGD	Mean Gamma Deviance (Tweedie with $p = 2$) $\frac{1}{n} \sum_{i=1}^n 2 \left(\ln \left(\frac{y_{p(i)}}{y_{a(i)}} \right) + \frac{y_{a(i)}}{y_{p(i)}} - 1 \right)$
MAPE	Mean Absolute Percentage Error $\frac{100}{n} \sum_{i=1}^n \frac{ y_{a(i)} - y_{p(i)} }{y_{a(i)}}$	MTD	Mean Tweedie Deviance (Tweedie with $p = 1.5$) $\frac{1}{n} \sum_{i=1}^n 2 \left(\frac{y_{a(i)}^{2-p}}{(1-p)(2-p)} - \frac{y_{a(i)} y_{p(i)}^{1-p}}{1-p} + \frac{y_{p(i)}^{2-p}}{2-p} \right)$

Metric	Description	Metric	Description
MSLE	Mean Squared Logarithmic Error $\frac{1}{n} \sum_{i=1}^n [\ln(1 + y_{a(i)}) - \ln(1 + y_{p(i)})]^2$	D²	Coefficient of determination for Tweedie deviance $D^2 = 1 - \frac{D(y_a, y_p)}{D(y_a, \bar{y}_a)}$ where $\bar{y}_a$ is the mean of y_a
MedAE	Median Absolute Error $\text{median}(y_{a(1)} - y_{p(1)} , \dots, y_{a(n)} - y_{p(n)})$	MPL	Mean Pinball Loss (quantile loss) with parameter $\alpha \in (0, 1)$ $L_\alpha = \frac{1}{n} \sum_{i=1}^n \begin{cases} \alpha(y_{a(i)} - y_{p(i)}), & y_{a(i)} \geq y_{p(i)} \\ (1 - \alpha)(y_{p(i)} - y_{a(i)}), & y_{a(i)} < y_{p(i)} \end{cases}$

* $y_{p(i)}$: Predicted value for i -th sample, $y_{a(i)}$: Actual value for i -th sample, n : number of samples, $\alpha = 0.5$ for MPL.

2. Prediction Performance using Scalar Metrics

Performance metrics of each model were obtained using the mean values (with confidence values) of 30 runs for each of the safety factors, F_s , F_o , and F_{ss} . The ten evaluation metrics for the model were assessed using their rank values [mishra2021probabilistic]. The models were ranked by performance, and the best model received a rank value of $R = 1$. Performance evaluation metrics and the corresponding rank R values are given in Tables 3, 4, and 5 for F_s , Tables 6, 7, and 8 for F_o , and Tables 9, 10 and 11 for F_{ss} .

Table 3. Performance metrics for E1 in prediction F_s .

Metric	RR	R	kNN	R	BR	R	SVR	R	LR	R	PR	R
MaxE	2.447±0.150	6	0.653±0.102	2	2.435±0.150	5	2.245±0.166	3	2.424±0.149	4	0.095±0.011	1
MAE	0.366±0.008	4	0.051±0.004	2	0.366±0.008	5	0.254±0.010	3	0.366±0.008	6	0.016±0.002	1
MSLE	0.033±0.001	4	0.001±0.000	2	0.033±0.001	5	0.018±0.001	3	0.033±0.001	6	0.000±0.000	1
MedAE	0.291±0.004	6	0.013±0.001	2	0.290±0.004	5	0.159±0.005	3	0.290±0.004	4	0.012±0.002	1
MAPE	0.485±0.022	4	0.028±0.001	2	0.486±0.022	5	0.258±0.005	3	0.487±0.022	6	0.025±0.003	1
MPD	0.174±0.005	4	0.004±0.001	2	0.174±0.005	5	0.105±0.007	3	0.176±0.006	6	0.001±0.000	1
MGD	1.834±1.664	5	0.002±0.000	1	1.702±1.354	5	4.790±8.743	6	1.519±0.917	4	0.002±0.001	2

Metric	RR	R	kNN	R	BR	R	SVR	R	LR	R	PR	R
MTD	0.256±0.021	6	0.014±0.003	2	0.255±0.021	5	0.169±0.020	3	0.255±0.021	4	0.001±0.000	1
D2	0.784±0.005	6	0.996±0.002	2	0.784±0.005	5	0.879±0.008	3	0.783±0.005	4	0.999±0.000	1
MPL	0.183±0.004	4	0.026±0.002	2	0.183±0.004	5	0.127±0.005	3	0.183±0.004	6	0.008±0.001	1
Total Rank		49		19		50		33		50		11

Table 4. Performance metrics for E2 in prediction F_s .

Metric	RR	R	kNN	R	BR	R	SVR	R	LR	R	PR	R
MaxE	1.296±0.081	6	0.748±0.043	2	1.284±0.081	5	1.066±0.071	3	1.282±0.081	4	0.041±0.004	1
MAE	0.209±0.004	6	0.086±0.002	2	0.208±0.004	4	0.100±0.003	3	0.208±0.004	5	0.005±0.001	1
MSLE	0.015±0.000	4	0.002±0.000	2	0.015±0.000	5	0.003±0.000	3	0.015±0.000	6	0.000±0.000	1
MedAE	0.179±0.004	6	0.042±0.001	2	0.179±0.004	5	0.067±0.001	3	0.178±0.004	4	0.004±0.000	1
MAPE	0.257±0.005	4	0.067±0.001	2	0.257±0.004	5	0.097±0.002	3	0.257±0.005	6	0.006±0.001	1
MPD	0.102±0.006	6	0.010±0.001	2	0.100±0.004	5	0.015±0.001	3	0.099±0.004	4	0.000±0.000	1
MGD	20.652±39.269	6	0.008±0.000	2	1.995±1.427	5	0.020±0.001	3	1.451±0.524	4	0.000±0.000	1
MTD	0.073±0.004	6	0.021±0.001	2	0.073±0.004	4	0.027±0.002	3	0.073±0.004	5	0.000±0.000	1
D2	0.890±0.004	6	0.970±0.002	2	0.890±0.004	4	0.965±0.002	3	0.890±0.004	5	0.999±0.000	1
MPL	0.104±0.002	6	0.043±0.001	2	0.104±0.002	4	0.050±0.001	3	0.104±0.002	5	0.002±0.000	1
Total Rank		56		20		46		30		48		10

Table 5. Performance metrics for E3 in prediction F_s .

Metric	RR	R	kNN	R	BR	R	SVR	R	LR	R	PR	R
MaxE	2.535±0.169	6	0.727±0.056	2	2.534±0.169	5	2.163±0.147	3	2.528±0.169	4	0.109±0.011	1

Metric	RR	R	kNN	R	BR	R	SVR	R	LR	R	PR	R
MAE	0.278±0.004	4	0.059±0.002	2	0.278±0.004	5	0.124±0.004	3	0.278±0.004	6	0.011±0.000	1
MSLE	0.023±0.000	4	0.001±0.000	2	0.023±0.000	5	0.005±0.000	3	0.023±0.000	6	0.000±0.000	1
MedAE	0.223±0.002	6	0.023±0.001	2	0.223±0.002	5	0.063±0.001	3	0.222±0.002	4	0.008±0.000	1
MAPE	0.357±0.009	4	0.045±0.001	2	0.357±0.009	5	0.109±0.002	3	0.358±0.009	6	0.016±0.001	1
MPD	0.136±0.004	4	0.006±0.000	2	0.137±0.004	5	0.026±0.001	3	0.138±0.004	6	0.000±0.000	1
MDG	1.926±1.297	4	0.004±0.000	2	2.062±0.754	5	0.031±0.003	3	4.837±2.764	6	0.001±0.000	1
MTD	0.158±0.008	6	0.013±0.001	2	0.158±0.008	5	0.066±0.006	3	0.158±0.008	4	0.000±0.000	1
D2	0.842±0.004	5	0.989±0.001	2	0.842±0.004	4	0.956±0.004	3	0.841±0.004	6	0.999±0.000	1
MPL	0.139±0.002	4	0.030±0.001	2	0.139±0.002	5	0.062±0.002	3	0.139±0.002	6	0.005±0.000	1
Total Rank		47		20		49		30		54		10

Table 6. Performance metrics for E1 in prediction F_o .

Metric	RR	R	kNN	R	BR	R	SVR	R	LR	R	PR	R
MaxE	3.547±0.063	5	0.329±0.040	2	3.546±0.063	4	4.424±0.113	6	3.542±0.063	3	0.114±0.024	1
MAE	0.698±0.015	4	0.043±0.002	2	0.698±0.015	5	0.586±0.019	3	0.699±0.015	6	0.029±0.004	1
MSLE	0.057±0.003	4	0.000±0.000	1	0.057±0.003	5	0.027±0.001	3	0.058±0.003	6	0.000±0.000	2
MedAE	0.603±0.010	5	0.024±0.001	1	0.603±0.010	4	0.391±0.008	3	0.603±0.010	6	0.024±0.003	2
MAPE	0.520±0.029	4	0.013±0.000	1	0.520±0.029	6	0.286±0.007	3	0.522±0.029	6	0.026±0.003	2
MPD	0.277±0.011	4	0.001±0.000	1	0.277±0.011	5	0.224±0.010	3	0.278±0.011	6	0.001±0.001	2
MGD	0.416±0.111	5	0.000±0.000	1	0.412±0.106	4	6.808±5.060	7	0.427±0.107	6	0.004±0.002	2
MTD	0.853±0.048	4	0.005±0.001	2	0.853±0.048	5	0.826±0.073	3	0.855±0.047	6	0.002±0.001	1
D2	0.888±0.002	4	0.999±0.000	2	0.888±0.002	5	0.933±0.004	3	0.887±0.002	6	0.9999±0.000	1

Metric	RR	R	kNN	R	BR	R	SVR	R	LR	R	PR	R
MPL	0.349±0.007	4	0.022±0.001	2	0.349±0.007	5	0.293±0.009	3	0.349±0.007	6	0.015±0.002	1
Total Rank		43		15		48		37		57		15

Table 7. Performance metrics for E2 in prediction F_o .

Metric	RR	R	kNN	R	BR	R	SVR	R	LR	R	PR	R
MaxE	2.326±0.086	5	1.109±0.083	2	2.322±0.085	4	2.587±0.116	6	2.320±0.084	3	0.064±0.005	1
MAE	0.486±0.008	5	0.171±0.004	2	0.486±0.008	4	0.370±0.012	3	0.486±0.008	6	0.010±0.001	1
MSLE	0.027±0.001	6	0.003±0.000	2	0.027±0.001	5	0.013±0.001	3	0.027±0.001	4	0.000±0.000	1
MedAE	0.458±0.008	6	0.099±0.004	2	0.458±0.008	4	0.268±0.007	3	0.458±0.009	5	0.007±0.001	1
MAPE	0.230±0.005	4	0.058±0.001	2	0.230±0.005	5	0.151±0.004	3	0.230±0.005	6	0.006±0.001	1
MPD	0.180±0.013	6	0.017±0.001	2	0.180±0.013	5	0.090±0.005	3	0.177±0.011	4	0.000±0.000	1
MGD	2.363±2.176	6	0.007±0.000	2	1.559±0.711	5	0.475±0.375	3	1.182±0.374	4	0.000±0.000	1
MTD	0.351±0.016	6	0.068±0.003	2	0.351±0.015	5	0.297±0.022	3	0.351±0.015	4	0.000±0.000	1
D2	0.921±0.002	6	0.986±0.001	2	0.921±0.002	5	0.957±0.002	3	0.921±0.002	4	1.000±0.000	1
MPL	0.243±0.004	5	0.086±0.002	2	0.243±0.004	4	0.185±0.006	3	0.243±0.004	6	0.005±0.001	1
Total Rank		55		20		46		33		46		10

Table 8. Performance metrics for E3 in prediction F_o .

Metric	RR	R	kNN	R	BR	R	SVR	R	LR	R	PR	R
MaxE	3.653±0.060	6	1.177±0.081	2	3.648±0.060	5	3.481±0.088	3	3.638±0.061	4	0.013±0.001	1
MAE	0.584±0.007	4	0.103±0.002	2	0.584±0.007	5	0.318±0.008	3	0.584±0.007	6	0.002±0.000	1
MSLE	0.043±0.002	4	0.002±0.000	2	0.043±0.002	5	0.011±0.000	3	0.043±0.002	6	0.000±0.000	1

Metric	RR	R	kNN	R	BR	R	SVR	R	LR	R	PR	R
MedAE	0.516±0.005	6	0.029±0.001	2	0.516±0.005	5	0.193±0.004	3	0.515±0.005	4	0.002±0.000	1
MAPE	0.387±0.015	5	0.035±0.001	2	0.387±0.015	6	0.145±0.003	3	0.387±0.015	4	0.002±0.000	1
MPD	0.234±0.009	4	0.010±0.000	2	0.236±0.009	5	0.107±0.005	3	0.237±0.009	6	0.000±0.000	1
MGD	1.172±0.380	3	0.004±0.000	2	2.832±1.646	6	2.014±1.091	4	2.533±1.897	5	0.000±0.000	1
MTD	0.571±0.021	5	0.040±0.002	2	0.571±0.021	4	0.288±0.019	3	0.572±0.021	6	0.000±0.000	1
D2	0.901±0.002	6	0.995±0.000	2	0.901±0.002	4	0.966±0.002	3	0.902±0.002	5	1.000±0.000	1
MPL	0.292±0.003	4	0.051±0.001	2	0.292±0.003	5	0.159±0.004	3	0.292±0.004	6	0.001±0.000	1
Total Rank		47		20		50		31		52		10

Table 9. Performance metrics for E1 in prediction F_{ss} .

Metric	RR	R	kNN	R	BR	R	SVR	R	LR	R	PR	R
MaxE	0.234±0.007	6	0.069±0.006	2	0.233±0.007	4	0.167±0.006	3	0.233±0.007	5	0.049±0.005	1
MAE	0.056±0.001	5	0.009±0.000	2	0.056±0.001	4	0.052±0.001	3	0.056±0.001	6	0.006±0.001	1
MSLE	0.001±0.000	4	0.000±0.000	2	0.001±0.000	5	0.001±0.000	3	0.001±0.000	6	0.000±0.000	1
MedAE	0.044±0.001	5	0.005±0.000	2	0.044±0.001	4	0.047±0.001	6	0.044±0.001	3	0.004±0.001	1
MAPE	0.041±0.001	4	0.006±0.000	2	0.041±0.001	5	0.037±0.001	3	0.041±0.001	6	0.004±0.001	1
MPD	0.004±0.000	4	0.000±0.000	2	0.004±0.000	5	0.003±0.000	3	0.004±0.000	6	0.000±0.000	1
MGD	0.004±0.000	4	0.000±0.000	2	0.004±0.000	5	0.002±0.000	3	0.004±0.000	6	0.000±0.000	1
MTD	0.005±0.000	6	0.000±0.000	2	0.005±0.000	4	0.004±0.000	3	0.005±0.000	5	0.000±0.000	1
D2	0.979±0.001	4	0.999±0.000	2	0.979±0.001	4	0.986±0.000	3	0.979±0.001	4	0.999±0.000	1
MPL	0.028±0.001	5	0.005±0.000	2	0.028±0.001	4	0.026±0.000	3	0.028±0.001	6	0.003±0.000	1
Total Rank		47		20		44		33		53		10

Table 10. Performance metrics for E2 in prediction F_{ss} .

Metric	RR	R	kNN	R	BR	R	SVR	R	LR	R	PR	R
MaxE	0.300±0.022	4	0.275±0.030	3	0.300±0.022	5	0.255±0.037	1	0.300±0.022	6	0.273±0.033	2
MAE	0.044±0.001	4	0.044±0.001	5	0.044±0.001	2	0.054±0.001	6	0.044±0.001	3	0.023±0.007	1
MSLE	0.001±0.000	3	0.001±0.000	2	0.001±0.000	4	0.001±0.000	6	0.001±0.000	5	0.000±0.000	1
MedAE	0.035±0.001	5	0.033±0.001	2	0.034±0.001	4	0.050±0.001	6	0.034±0.001	3	0.013±0.006	1
MAPE	0.033±0.001	3	0.030±0.001	2	0.033±0.001	4	0.039±0.001	6	0.033±0.001	5	0.017±0.005	1
MPD	0.003±0.000	3	0.002±0.000	2	0.003±0.000	4	0.003±0.000	6	0.003±0.000	5	0.002±0.001	1
MGD	0.002±0.000	3	0.002±0.000	2	0.002±0.000	4	0.003±0.000	6	0.002±0.000	5	0.001±0.001	1
MTD	0.004±0.000	4	0.004±0.000	5	0.004±0.000	2	0.004±0.000	6	0.004±0.000	3	0.002±0.002	1

Metric	RR	R	kNN	R	BR	R	SVR	R	LR	R	PR	R
D2	0.980±0.001	3	0.981±0.001	2	0.980±0.001	3	0.980±0.001	5	0.980±0.001	3	0.992±0.007	1
MPL	0.022±0.000	4	0.022±0.000	5	0.022±0.000	2	0.027±0.000	6	0.022±0.000	3	0.012±0.003	1
Total Rank		36		30		34		54		41		11

Table 11. Performance metrics for E3 in prediction F_{ss} .

Metric	RR	R	kNN	R	BR	R	SVR	R	LR	R	PR	R
MaxE	0.312±0.019	4	0.289±0.033	3	0.312±0.019	5	0.264±0.034	1	0.313±0.019	6	0.269±0.022	2
MAE	0.049±0.001	3	0.031±0.001	2	0.049±0.001	4	0.052±0.001	6	0.049±0.001	5	0.013±0.000	1
MSLE	0.001±0.000	4	0.000±0.000	2	0.001±0.000	5	0.001±0.000	3	0.001±0.000	6	0.000±0.000	1
MedAE	0.038±0.001	4	0.020±0.001	2	0.038±0.001	5	0.048±0.001	6	0.038±0.001	3	0.006±0.000	1
MAPE	0.037±0.001	3	0.021±0.000	2	0.037±0.001	4	0.037±0.000	6	0.037±0.001	5	0.009±0.000	1
MPD	0.003±0.000	4	0.001±0.000	2	0.003±0.000	5	0.003±0.000	3	0.003±0.000	6	0.001±0.000	1
MGD	0.003±0.000	4	0.001±0.000	2	0.003±0.000	5	0.002±0.000	3	0.003±0.000	6	0.001±0.000	1
MTD	0.004±0.000	6	0.002±0.000	2	0.004±0.000	4	0.004±0.000	3	0.004±0.000	5	0.001±0.000	1
D2	0.984±0.001	5	0.992±0.000	2	0.984±0.001	4	0.988±0.001	3	0.984±0.001	6	0.997±0.000	1
MPL	0.025±0.000	3	0.015±0.000	2	0.025±0.000	4	0.026±0.000	6	0.025±0.000	5	0.006±0.000	1
Total Rank		40		21		45		40		53		11

3. Selected PR models for Cantilever Stability Assessment

Because the PR-based equations were very long and hard to apply, they were converted into functions that take X_1 , X_2 , X_3 , X_4 , and ϕ as input parameters for user convenience. The functions are provided in the Jupyter Notebook file

"RetainingWallSafetyCheck_LibraryManual.ipynb" shared on GitHub.

Equation 1: Sliding safety factor PR model calibrated with E3 Dataset

$$\begin{aligned} F_s = & -0.0027 + X_1(-2.4604) + X_2(1.9860) + X_3(-30.8609) + X_4(10.9658) + \phi(-1.8574) + X_1^2(-0.0136) + X_1X_2(0.0227) + X_1X_3 \\ & + X_1X_4(-1.1160) + X_1\phi(0.2275) + X_2^2(-0.0261) + X_2X_3(-2.6140) + X_2X_4(0.9195) + X_2\phi(-0.1848) + X_3^2(23.2641) + X_3X_4 \\ & + X_4\phi(-0.9356) + \phi^2(0.0722) + X_1^3(0.0051) + X_1^2X_2(-0.0189) + X_1^2X_3(-0.0144) + X_1^2X_4(-0.0493) + X_1^2\phi(-0.0002) + X_1X_2^2 \\ & + X_1X_2\phi(0.0013) + X_1X_3^2(-1.7552) + X_1X_3X_4(2.1453) + X_1X_3\phi(-0.1910) + X_1X_4^2(-27.8230) + X_1X_4\phi(0.0787) + X_1\phi^2(-0.0001) \\ & + X_2^2\phi(-0.0017) + X_2X_3^2(1.3112) + X_2X_3X_4(-1.3922) + X_2X_3\phi(0.1481) + X_2X_4^2(31.2189) + X_2X_4\phi(-0.0725) + X_2\phi^2(0.005) \\ & + X_3X_4^2(-69.0748) + X_3X_4\phi(0.9039) + X_3\phi^2(-0.0571) + X_4^3(16.7722) + X_4^2\phi(-0.6290) + X_4\phi^2(0.0210) + \phi^3(-0.0012) + X_1^4 \\ & + X_1^3X_4(0.0014) + X_1^3\phi(-0.0000) + X_1^2X_2^2(0.0024) + X_1^2X_2X_3(0.0004) + X_1^2X_2X_4(-0.0170) + X_1^2X_2\phi(0.0001) + X_1^2X_3^2(0.001) \\ & + X_1^2X_4\phi(0.0003) + X_1^2\phi^2(0.0000) + X_1X_2^3(-0.0096) + X_1X_2^2X_3(0.0055) + X_1X_2^2X_4(0.0768) + X_1X_2^2\phi(-0.0001) + X_1X_2X_3^2(0.001) \\ & + X_1X_2X_4\phi(-0.0029) + X_1X_3^3(0.3889) + X_1X_3^2X_4(-0.2856) + X_1X_3^2\phi(0.0430) + X_1X_3X_4^2(5.3755) + X_1X_3X_4\phi(-0.0679) + X_1X_3\phi^2(-0.0001) \\ & + X_1\phi^3(0.0001) + X_2^4(0.0078) + X_2^3X_3(-0.0097) + X_2^3X_4(-0.0602) + X_2^3\phi(-0.0000) + X_2^2X_3^2(-0.0080) + X_2^2X_3X_4(0.1394) + \\ & + X_2^2\phi^2(0.0000) + X_2X_3^3(-0.2970) + X_2X_3^2X_4(0.2936) + X_2X_3^2\phi(-0.0327) + X_2X_3X_4^2(-1.8963) + X_2X_3X_4\phi(0.0529) + X_2X_3\phi^2(-0.0001) \\ & + X_2\phi^3(-0.0001) + X_3^4(2.7201) + X_3^3X_4(6.3210) + X_3^3\phi(0.1296) + X_3^2X_4^2(30.2769) + X_3^2X_4\phi(-0.2348) + X_3^2\phi^2(0.0113) + X_3X_4^3(0.0001) \\ & + X_3\phi^3(0.0005) + X_4^4(1.3667) + X_4^3\phi(-6.0400) + X_4^2\phi^2(0.0067) + X_4\phi^3(-0.0002) + \phi^4(0.0000) + 17.6615 \end{aligned}$$

Equation 2: Overturning safety factor PR model calibrated with E3 Dataset

$$\begin{aligned}
F_o = & X_1X_2\phi(0.0002) + X_1X_3^2(-0.0559) + X_1X_3X_4(-0.0469) + X_1X_3\phi(0.0241) + X_1X_4^2(-5.8990) + X_1X_4\phi(0.0107) + X_1\phi^2(-0.0003) \\
& + X_2^2X_3(0.0236) + X_2^2X_4(-0.0433) + X_2^2\phi(0.0020) + X_2X_3^2(0.0541) + X_2X_3X_4(-0.0582) + X_2X_3\phi(-0.0172) + X_2X_4^2(2.9709) \\
& + X_3^2X_4(-2.3647) + X_3^2\phi(-0.0087) + X_3X_4^2(4.5902) + X_3X_4\phi(-0.0584) + X_3\phi^2(0.0018) + X_4^3(5.6591) + X_4^2\phi(0.7788) + X_4\phi \\
& + X_1^4(0.0000) + X_1^3X_2(0.0000) + X_1^3X_3(-0.0002) + X_1^3X_4(0.0019) + X_1^3\phi(0.0000) + X_1^2X_2^2(0.0011) + X_1^2X_2X_3(0.0006) + X_1^2X_2\phi \\
& + X_1^2X_3X_4(-0.0037) + X_1^2X_3\phi(-0.0023) + X_1^2X_4^2(-0.0687) + X_1^2X_4\phi(-0.0012) + X_1^2\phi^2(0.0002) + X_1X_2^3(-0.0034) + X_1X_2^2X_3 \\
& + X_1X_2X_3^2(0.0036) + X_1X_2X_3X_4(0.0045) + X_1X_2X_3\phi(0.0001) + X_1X_2X_4^2(0.1418) + X_1X_2X_4\phi(-0.0005) + X_1X_2\phi^2(0.0000) \\
& + X_1X_3X_4^2(0.4218) + X_1X_3X_4\phi(0.0022) + X_1X_3\phi^2(-0.0005) + X_1X_4^3(96.3990) + X_1X_4^2\phi(0.0017) + X_1X_4\phi^2(-0.0000) + X_1\phi \\
& + X_2^3X_3(0.0005) + X_2^3X_4(-0.0097) + X_2^3\phi(-0.0000) + X_2^2X_3^2(-0.0321) + X_2^2X_3X_4(0.0054) + X_2^2X_3\phi(0.0036) + X_2^2X_4^2(0.1726) \\
& + X_2X_3^3(0.0310) + X_2X_3^2X_4(-0.0077) + X_2X_3^2\phi(-0.0036) + X_2X_3X_4^2(-1.0184) + X_2X_3X_4\phi(0.0137) + X_2X_3\phi^2(0.0003) + X_2 \\
& + X_3^4(-0.0255) + X_3^3X_4(1.2028) + X_3^3\phi(-0.0004) + X_3^2X_4^2(-2.9096) + X_3^2X_4\phi(0.0036) + X_3^2\phi^2(0.0001) + X_3X_4^3(2.8220) + X_3 \\
& + X_4^4(0.4892) + X_4^3\phi(-8.0371) + X_4^2\phi^2(-0.0048) + X_4\phi^3(-0.0000) + \phi^4(0.0000) + 1.9629
\end{aligned}$$

Equation 3: Slope stability safety factor PR model calibrated with E3 Dataset

$$\begin{aligned}
F_{ss} = & (-0.0045) + X_1(-0.2033) + X_2(0.2968) + X_3(-6.3327) + X_4(-1.3367) + \phi(-0.6050) + X_1^2(0.0438) + X_1X_2(0.0638) \\
& + X_1X_3(-0.0395) + X_1X_4(3.4060) + X_1\phi(0.0132) + X_2^2(-0.0564) + X_2X_3(-0.8630) + X_2X_4(1.0067) + X_2\phi(-0.0208) + X_3^2 \\
& + X_3X_4(24.1393) + X_3\phi(0.0772) + X_4^2(-338.4436) + X_4\phi(-0.2406) + \phi^2(0.0312) + X_1^3(-0.0182) + X_1^2X_2(-0.0044) + X_1^2X_3 \\
& + X_1^2X_4(0.2901) + X_1^2\phi(0.0027) + X_1X_2^2(0.0326) + X_1X_2X_3(0.1263) + X_1X_2X_4(-0.6146) + X_1X_2\phi(-0.0067) + X_1X_3^2(0.019 \\
& + X_1X_3\phi(-0.0066) + X_1X_4^2(-63.5125) + X_1X_4\phi(-0.0535) + X_1\phi^2(-0.0004) + X_2^3(-0.0378) + X_2^2X_3(-0.0605) + X_2^2X_4(-0 \\
& + X_2X_3^2(0.2496) + X_2X_3X_4(4.4673) + X_2X_3\phi(0.0281) + X_2X_4^2(-39.0857) + X_2X_4\phi(-0.0360) + X_2\phi^2(0.0005) + X_3^3(-18.466 \\
& + X_3^2\phi(0.0311) + X_3X_4^2(118.8547) + X_3X_4\phi(-0.5822) + X_3\phi^2(-0.0031) + X_4^3(40.9212) + X_4^2\phi(25.7036) + X_4\phi^2(-0.0010) + \\
& + X_1^4(0.0013) + X_1^3X_2(0.0080) + X_1^3X_3(-0.0096) + X_1^3X_4(-0.0795) + X_1^3\phi(-0.0001) + X_1^2X_2^2(-0.0414) + X_1^2X_2X_3(0.0042) \\
& + X_1^2X_2\phi(0.0002) + X_1^2X_3^2(0.0369) + X_1^2X_3X_4(0.6945) + X_1^2X_3\phi(0.0005) + X_1^2X_4^2(-0.7944) + X_1^2X_4\phi(0.0058) + X_1^2\phi^2(-0.00 \\
& + X_1X_2^2X_3(0.0301) + X_1X_2^2X_4(-0.1145) + X_1X_2^2\phi(-0.0002) + X_1X_2X_3^2(-0.1942) + X_1X_2X_3X_4(-0.5530) + X_1X_2X_3\phi(-0.0 \\
& + X_1X_2\phi^2(0.0001) + X_1X_3^3(-0.0819) + X_1X_3^2X_4(2.9972) + X_1X_3^2\phi(0.0007) + X_1X_3X_4^2(-31.0301) + X_1X_3X_4\phi(0.0240) + X_1 \\
& + X_1X_4^2\phi(0.5949) + X_1X_4\phi^2(-0.0001) + X_1\phi^3(0.0000) + X_2^4(-0.0412) + X_2^3X_3(-0.0396) + X_2^3X_4(0.1997) + X_2^3\phi(0.0002) + \\
& + X_2^2X_3X_4(-0.0264) + X_2^2X_3\phi(-0.0018) + X_2^2X_4^2(-18.6572) + X_2^2X_4\phi(0.0114) + X_2^2\phi^2(-0.0001) + X_2X_3^3(0.1880) + X_2X_3^2X_4 \\
& + X_2X_3X_4^2(-36.7587) + X_2X_3X_4\phi(0.0979) + X_2X_3\phi^2(-0.0002) + X_2X_4^3(55.2128) + X_2X_4^2\phi(1.2445) + X_2X_4\phi^2(0.0004) + X_2 \\
& + X_3^3X_4(-5.9986) + X_3^3\phi(-0.0073) + X_3^2X_4^2(175.4480) + X_3^2X_4\phi(0.0027) + X_3^2\phi^2(-0.0003) + X_3X_4^3(42.5366) + X_3X_4^2\phi(-5. \\
& + X_3\phi^3(0.0000) + X_4^4(4.0055) + X_4^3\phi(-29.2032) + X_4^2\phi^2(-0.4011) + X_4\phi^3(0.0001) + \phi^4(0.0000) + (5.5167)
\end{aligned}$$

4. Further Model Verification with Unseen Data

To assess the veracity and robustness of these models, an additional set of analyses was carried out using unseen data—i.e., data that were not part of any of the training and testing datasets (E1, E2, or E3) utilized in model development. Table 12 displays the numerical values of the unseen data, along with the safety factors computed using both the actual mathematical models and the PR models selected.

Table 12. Unseen data set with predicted and actual safety factors.

No	X_{1u}	X_{2u}	X_{3u}	X_{4u}	ϕ_u	Actual			Model Predictions		
						F_s	F_o	F_{ss}	F_s	F_o	F_{ss}
20	5.76	1.61	0.80	0.018	34	2.45	7.64	2.07	2.4340	7.4873	2.0551

21	2.16	0.72	0.48	0.032	27	0.44	0.85	1.11	0.4458	0.8435	1.0987
22	3.00	1.68	0.82	0.014	38	1.36	2.02	1.86	1.3472	1.9463	1.8406
23	4.62	2.50	0.70	0.028	23	0.52	2.59	1.07	0.5194	2.5744	1.0506
24	3.36	1.50	0.58	0.033	34	1.06	2.41	1.59	1.0718	2.3960	1.6013
25	4.92	2.90	0.82	0.038	30	0.96	3.56	1.49	0.9487	3.5101	1.4662
26	2.58	1.50	0.45	0.024	39	0.96	1.58	1.70	0.9568	1.5648	1.6862
27	3.20	0.80	0.58	0.005	20	0.40	1.44	0.91	0.4014	1.4348	0.9223
28	4.70	0.60	0.70	0.032	34	2.24	5.46	2.01	2.2193	5.3774	1.9820
29	4.50	1.45	0.38	0.013	28	1.00	3.88	1.38	1.0101	3.8735	1.3627
30	3.90	0.90	0.60	0.023	33	1.50	3.57	1.74	1.4912	3.5345	1.7337

5. Model input-output correlations

The resulting correlation coefficients and factors of safety are illustrated in Figure 1. Results reveal that X_1/H (the overall aspect ratio of the CRW) and X_3/H (the base thickness ratio) are, respectively, the most and least significant elements influencing stability. Furthermore, it was noted that 48% of the F_s values among the 2275 designs exceeded 1.1, 45% fell below 0.9, and 7% ranged between 0.9 and 1.1. For F_o , the respective percentages were 76%, 19%, and 5%. Ultimately, for F_{ss} , the ratios were 71%, 8%, and 21%. Upon analyzing these hazardous designs (i.e., $F_x \geq 1.10$), it was noted that the predominant failure mechanism is sliding. In all instances of failure, the mean values of X_1/H , X_2/X_1 , X_b/X_1 , and X_3/H were 0.70, 0.35, 0.50, and 0.10, respectively.

Figure 1. Effect graph of failure scenarios for wall dimensions.

 No description has been provided for this image

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